

ABSTRACT

The Department Library Management System plays a critical role in the efficient organization and management of library resources, ensuring seamless operations and enhancing user experience. This project aims to design and implement a comprehensive Library Management System that streamlines the management of books, users, and transactions within a library. The system will include features such as providing administrative functionalities such as user registration, managing library resources and fine management. It Allows students to search for books, view book details, reserve and borrow books, renew borrowed items. It also allows organizing library resources into categories, subjects, authors, and topics for easy navigation and retrieval. The project provides a user-friendly interface, developed with modern tools and frameworks, enhances accessibility for both library staff and students. By automating routine tasks and reducing manual errors, the Library Management System optimizes resource utilization and facilitates efficient library operations. This project not only demonstrates the application of software engineering principles but also underscores the importance of digital solutions in modernizing traditional library systems.

CHAPTER 1

INTRODUCTION

Library management has been central to human progress, adapting to the evolving needs of society over centuries. From primitive records on clay tablets to the advanced digital systems of today, libraries have continually developed new ways to organize and disseminate information. This evolution mirrors technological advancements, cultural shifts, and changing societal expectations, showcasing the enduring role of libraries as vital institutions of knowledge preservation.

The invention of the printing press in the 15th century marked a pivotal moment, making books more affordable and accessible. This technological advancement enabled libraries to expand their collections significantly, shifting the librarian's role from copying manuscripts to managing acquisitions and organization. National libraries also began to take shape, serving as repositories for cultural and historical artifacts. Figures like Sir Antonio Panizzi, who developed innovative cataloging methods and advocated for public access, exemplified the growing complexity of library management during this time.

By the 19th and 20th centuries, library management became increasingly professionalized. The establishment of organizations such as the American Library Association and the creation of formal library science programs helped standardize practices and equip librarians with the necessary skills to manage expanding collections. Libraries transitioned from static repositories to dynamic public institutions, offering educational programs, reference services, and outreach initiatives like traveling libraries. These efforts made knowledge more accessible to diverse communities, including those in rural or underserved areas.

A Library Management System has three basic functions to perform inside a library, referred by Sarma : 'Housekeeping Operations', 'Public Services', and 'Administrative Planning or Decision Making'. Housekeeping operations include acquisition of documents, technical processing like classification and cataloguing, circulation control, etc. Public services include online public access catalogue (OPAC), and Administrative planning or decision making includes formation of transaction policies, customisation of LMS etc.[1]

The Department Library Management System is a software solution developed to modernize and streamline the management of library resources within our department. Digitization of library helps to keep all the records secured and retrievable which saves the paperwork and made library

records easy to store. This project not only helps to store the library records but it also provides access to the library management staffs and students to check the information related to the books. Traditionally, maintaining library records involved manual processes, which are not only time-consuming but also prone to errors like misplaced books, duplicate records, or delayed updates. This system replaces these outdated methods with an efficient, automated approach, allowing for smooth management of books, student and faculty records, and transactions. With features like adding, updating, and searching for books, it ensures that library operations are quicker and more accurate, reducing the workload on staff while improving overall organization. In addition to basic management functions, the system introduces tracking for book issuance and returns, ensuring users have timely access to the resources they need while avoiding penalties for overdue books. Inventory management is also simplified, allowing staff to monitor book availability and replenish resources when needed. The user-friendly interface makes it easy for anyone to navigate the system, whether they are library staff, students, or faculty. By improving efficiency and minimizing errors, this system not only enhances the library's day-to-day operations but also promotes a better experience for everyone, creating a well-organized and accessible resource hub for the department.

1.1 OBJECTIVE

- The primary objective of the Library Management System project is to design and develop an efficient and secure software application that automates and streamlines the operations of a library. .
- The system aims to enhance accessibility and convenience for users by providing advanced search functionalities and minimizing manual errors and administrative workload through automation of routine tasks.
- Developing an intuitive and user-friendly interface for library staff and patrons to facilitate easy navigation
- bridges the gap between traditional library operations and modern digital solutions, promoting efficient resource management and improved user experience.
- adopts an organized and systematic approach to library resource management.

1.2 SCOPE

The Library Management System project has a broad and impactful scope, aiming to digitize and optimize library operations for improved efficiency and user satisfaction. The system will be designed to cater to the needs of both library administrators and users. For administrators, it provides tools to manage book inventory, monitor borrowing trends, handle overdue items, and generate reports for better decision-making. For users, the system offers a streamlined experience with features such as book search by title, author, or genre, real-time availability updates, and seamless borrowing and returning processes. The project has potential for future enhancements, such as integrating e-books, online reservation systems, and support for mobile applications, thereby making the library accessible anytime and anywhere. This project serves as a practical implementation of software engineering concepts and database management systems, addressing real-world challenges faced by traditional libraries and paving the way for modern and efficient library operations.

CHAPTER 2

LITERATURE SURVEY

2.1 EVOLUTION OF LIBRARY MANAGEMENT SYSTEMS: PAST, PRESENT AND FUTURE

Past: The development of library management dates back to ancient civilizations, where libraries were primarily focused on organizing and preserving knowledge. Early examples from Mesopotamia and Egypt featured libraries as places for spiritual and intellectual enrichment. As time progressed, especially in Ancient Greece and Rome, libraries developed systems for organization and management. The invention of the printing press in the 15th century further influenced the growth of libraries, making books more accessible and shifting the role of librarians.

Modern: library management has evolved into a complex, strategic discipline influenced by technological advancements and changing user expectations. Modern libraries have transitioned from being passive repositories to proactive service providers, adopting business management concepts such as strategic planning, customer service, and cost analysis. Professional associations like the American Library Association (ALA) and the Library Leadership and Management Association (LLAMA) provide support and resources for library managers, while library education has become more specialized, with degrees like MLS and MLIS becoming standard. The integration of information technology (IT) has transformed library operations, with the automation of business functions, the creation of online catalogs, and the adoption of electronic databases. However, libraries now face the challenge of balancing physical and digital collections, as well as addressing issues like copyright, digital preservation, and interoperability.

Future: Looking ahead, library management faces challenges and opportunities brought about by digital transformation. Libraries are increasingly focused on creating digital collections, collaborating globally, and embracing new technologies to meet user expectations. However, they must navigate issues like digital preservation, copyright, and system interoperability. The future of libraries will require strong leadership, strategic innovation, and a balance between maintaining physical collections and developing new digital resources to ensure continued relevance in an increasingly digital world.[2]

2.2 FUNCTIONAL CAPABILITIES OF LIBRARY MANAGEMENT SYSTEMS

Library Management System has been intended to automate, oversee and care for the general handling of even enormous scale libraries. This product is fit for overseeing Book Issues, Returns, Calculating/Managing various data, creating different Reports for Record-Keeping and Review purposes, as per end client prerequisites.

it also needs to keep a thorough database of the members, the Management System stores each user's name, ID, and password for aiding in determining the member's history. The software is user-friendly and improves the effectiveness of the librarian and library administration.

A modern integrated library management system (LMS). Can be scalable to Windows, Linux and Mac OS platform, Full catalog, circulation and acquisitions system for library stock management, Simple, clear search interface for all users, Multilingual and multi-user support, Export and import records, Easy way to enter new books, Keep record of complete info of a book like title, author name, publisher name etc, Easy way to check-in and check-out, It is user-friendly software, Keep record of different categories, Classify the books subject wise, Easy to maintain database,[4]

2.3 ONLINE LIBRARY MANAGEMENT SYSTEM FOR ENHANCED ACCESSIBILITY AND EFFICIENCY

Online Library Management System is an Automated Library System that handles the various functions of the library. It provides a complete solution to the library management software. The online Library Management System is classified into two parts: Bar Code System and RFID System.

Library plays an important role in all schools and colleges, no educational institution can exist without Library Administration Software. It is an important part of every school and college and it helps the librarian to keep records of available books as well as issued books. Library Management System software helps in different ways by providing students the facility to learn, gather resources, promote group learning and improve knowledge and skills.

The demand for Library Automation Software in India is increasing day by day with the advancement of the system. As we all are living in the digital age, everyone prefers quick and accurate service. Almost all schools and colleges prefer the Library Management System due

to the organized and systematic management of the database. College Library Software helps the students to prepare their projects easily and in less time. It also helps in preparing notes and assignments.

Library Management System helps the students in better learning by providing quick access to the library system. With Advance Library Software they can easily find books, catalogs, magazines of their interest. The library is a great place to study and make notes which ultimately help in scoring a good score.

2.4 KEY CHALLENGES AND LIMITATIONS IN LIBRARY MANAGEMENT SYSTEMS

Library management systems (LMS) face several challenges related to user experience. A complex or non-intuitive user interface (UI) can make it difficult for both librarians and patrons to navigate the system effectively. Furthermore, performance issues, especially in large libraries with extensive collections, can hinder the system's efficiency, particularly during peak hours. Many systems also struggle with scalability, making it difficult to handle growing collections and user bases. Effective search functionality is another key issue, with many systems lacking intuitive search options, hindering users' ability to quickly locate materials. Poor reporting tools also present problems, as limited or poorly designed reporting capabilities prevent library staff from making data-driven decisions.

The integration capabilities of LMS are often limited, restricting the ability to connect the system with other important tools like accounting software or e-learning platforms. Additionally, data migration can become problematic when transitioning from one system to another, with poorly designed migration tools leading to data loss or corruption. Many systems also fail to provide sufficient automation for routine tasks such as sending overdue notices or performing inventory checks, adding unnecessary manual work for library staff. Mobile accessibility is another area where many LMS solutions fall short, making it difficult for patrons to access services via smartphones or tablets. Moreover, the inflexibility of lending rules in many systems limits customization for libraries with unique policies.

Security vulnerabilities remain a major concern for LMS, as insufficient protection can lead to unauthorized access to user data and library records. Limited customization options also hinder libraries from tailoring the system to meet their specific operational requirements, leading to inefficiency. Compatibility issues across different operating systems and devices can also limit

accessibility, forcing users to deal with inconsistent experiences across platforms. Furthermore, a lack of proper user training from LMS providers can lead to inefficiencies and user frustration, as libraries are left to figure out the system on their own. Inadequate technical support exacerbates this issue, leaving libraries without timely assistance when problems arise.

The cost of upgrading LMS software is another challenge, as many providers charge high fees for upgrades, creating financial and operational barriers for libraries. Additionally, inadequate backup and recovery options pose a significant risk to library data, especially in the event of hardware failures or cyberattacks. Licensing restrictions can further complicate library operations by limiting the number of users or installations allowed under the license, leading to additional costs as libraries grow. Many LMS solutions also struggle to manage digital resources, such as e-books and audiobooks, which are increasingly important in modern libraries. Finally, issues with user account management, such as complicated registration processes and password reset problems, frustrate both staff and patrons.[5]

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

The Software Requirements Specification (SRS) is a foundational document in the software development lifecycle, serving as a blueprint for the development, testing, and maintenance of a software system. It provides a detailed description of the system's intended behavior, outlining both the functional and non-functional requirements that define what the system must accomplish and how it should perform. The functional requirements describe the system's features, such as user interactions, inputs, outputs, and processes, while non-functional requirements focus on performance metrics, security, reliability, and usability. An SRS not only ensures that all stakeholders—developers, testers, project managers, and clients—share a common understanding of the system's scope and objectives but also helps in minimizing misunderstandings and ambiguities. Additionally, it acts as a reference point throughout the entire software development process, supporting design decisions, providing a basis for quality assurance, and guiding the future evolution or maintenance of the system. By documenting assumptions, constraints, and dependencies, the SRS helps anticipate potential risks and challenges. Ultimately, the SRS is an essential tool for ensuring that the final software product meets the user's needs and adheres to the agreed-upon specifications.

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)FOR DEPARTMENT LIBRARY MANAGEMENT SYSTEM

3.1 INTRODUCTION

3.1.1 PURPOSE

The purpose of the Library Management System (LMS) is to provide an efficient and user-friendly software solution for managing the day-to-day operations of a library. The system aims to automate and streamline processes such as book cataloging, user registration, borrowing and returning books. Ultimately, the system will aim to simplify the management of library resources while providing a seamless experience for all users.

3.1.2 SCOPE

The scope of this project includes the following key functionalities:

- Registration, authentication, and role-based access control for users
- Adding, updating, and removing books from the library catalog.
- Enabling users to search the catalog by various criteria like title, author.
- Recording all transactions, including borrowed and returned books and user activity.
- Providing insights to help manage the library effectively.

3.2 OVERALL DESCRIPTION

3.2.1 PRODUCT PERSPECTIVE

The Library Management System (LMS) is a standalone application designed to automate and enhance the core functionalities of library operations. It will serve as a centralized system for managing library resources, including books, users, and transactions. The system will be deployed on a web-based platform, making it accessible to both library staff and members from any device with internet access. The LMS will integrate seamlessly with a backend database to store and manage data on books, users, and transactions.

3.2.2 PRODUCT FUNCTIONS

- Database Management: Maintain a comprehensive database of library resources, including information about new acquisitions, borrowed items, and due dates.
- Automated Processes: Streamline tasks such as book indexing, cataloging, reservations, and notifications to users.
- Categorization: Organize library resources into categories, subjects, authors, and topics for easy navigation and retrieval.
- User Management: Manage library memberships, issue library cards, and handle user requests and inquiries.
- Admin Panel: Provides administrative functionalities such as user registration, managing library resources, generating reports, and maintaining records.

- **Student Panel:** Allows students to search for books, view book details, reserve and borrow books, renew borrowed items, and analyze their transaction history.
- **Faculty Staff Panel:** Similar to the student panel, tailored for faculty and staff members, enabling them to access library resources and perform relevant tasks.

3.2.3 USER CHARACTERISTICS

- **General Users:** General users are the members of the library who will use the system to search for books, borrow and return books, and manage their personal account information. Their primary interaction with the system will be related to managing their borrowing activity and viewing the library's catalog.
- **Admin Users:** Admin users, typically librarians or library staff, will have full access to the system's administrative functions. They will be responsible for managing the library's book catalog, including adding, updating, or removing books. Admin users will also manage user registrations, define roles, and generate reports on library activity.

3.3 FUNCTIONAL REQUIREMENTS

3.3.1 User Management

User Registration: Librarians can create, update, and deactivate patron accounts. Mandatory fields: name, contact info, and library card number.

Role-Based Access Control: Define roles (Admin, Patron). Admins manage system settings, Librarians handle transactions, Patrons access self-service features.

Patron Self-Service: Patrons can view checkouts, holds, fines, update contact info, and renew items via a secure portal.

3.3.2 Catalog Management

Add/Edit Catalog Entries: Librarians can add/update items (books, e-books, DVDs) with details (title, author, ISBN, format, copies).

Search and Browse: Users can search by title, author, genre, or keyword. Filters for availability, format, and publication year.

Inventory Tracking: Track item status (available, checked out, lost) and manage multiple copies.

3.3.3 Circulation Management

Check-Out Items: Scan item/user barcodes, assign due dates, and validate patron eligibility (no overdue items/fines).

Check-In Items: Automatically update status, calculate fines if overdue, and notify reserved patrons.

Renewals: Allow renewals (up to 3 times) if no holds exist.

3.3.4 Reservations & Holds

Place Holds: Patrons can reserve unavailable items. Notify when available and hold for 7 days before expiration.

Hold Queue Management: Librarians can view/modify hold queues and manually assign items.

3.3.5 Fines & Fees

Automatic Fines Calculation: Overdue fines will be collected on a daily basis, with the option for waivers subject to approval.

Payment Processing: Integrate payment gateways for online fines payment. Generate receipts.

3.3.6 Reporting & Analytics

Generate Reports: Overdue items, checkout history, popular items, fines collected. Export to PDF/Excel.

Real-Time Dashboard: Display daily checkouts, active holds, and overdue items for librarians.

3.3.7 Notifications

Automated Alerts: Email/SMS for due dates, overdue items, holds availability, and account changes. Customizable templates.

3.3.8 System Configuration

Manage Policies: Admins set loan periods (e.g., 14 days), fine rates, max checkouts (e.g., 10 items), and hold expiration.

3.3.9 Integration

Third-Party Integration: Support barcode scanners, email/SMS APIs.

3.3.10 Security & Compliance

Authentication: Secure login with password policies, and session timeout (15 minutes).

3.3.11 Backup & Recovery

Data Backup: Schedule daily backups and allow manual backups. Restore data from backups.

3.3.12 Additional Features

Bulk Import/Export: Import/export user and catalog data via CSV/Excel.

3.4 NON-FUNCTIONAL REQUIREMENTS

3.4.1 USABILITY

- The system shall have an intuitive and user-friendly interface suitable for users with basic computer literacy.

3.4.2 COMPATIBILITY

- The system shall be compatible with major web browsers.(e.g., Chrome, Firefox, Edge, Safari).

3.4.3 PORTABILITY

- The system shall be deployable on multiple operating systems (e.g., Windows, Linux, macOS).

3.4.4 AVAILABILITY

- The system will be available whenever necessary.

CHAPTER 4

SOFTWARE DESIGN DOCUMENTS

4.1 SYSTEM ARCHITECTURE

The Library Management System follows a 3-tier architecture to separate concerns and ensure modularity [6]

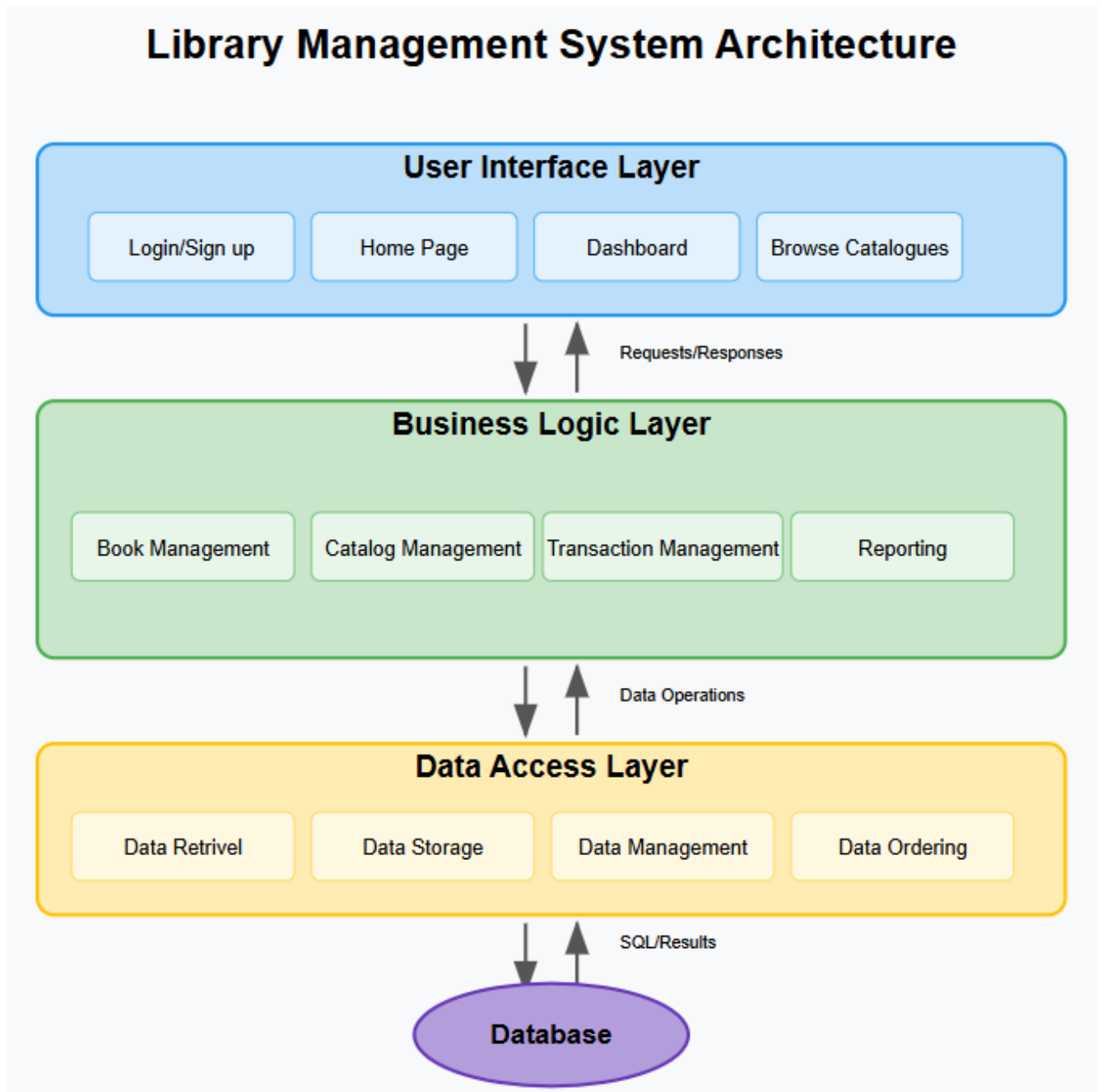


Figure 1: Three Tier architecture of library management system

4.1.1 User Interface Layer [Front-end]

This layer is responsible for interacting with the end-users of the system. It includes An interface for library staff and patrons to manage books, members, and transactions. The application is composed of a series of interactive web pages. Through this layer, users can directly enter the system to complete the required functions. According to the user category, different users would log in to the system with different identities. There are three types of this system, namely, students, teachers and librarian. Their permissions differ generally from low to high Teachers and students can only borrow, read and consult library archives. Differences exist between teachers and students in the scope and right of use. The librarian can sort out and optimize various resources of library management system, and gradually improve the level of user interface. The user layer is usually logged in as a browser, which is extremely convenient to use and has a friendly interface.

4.1.2 Business logic layer [Back-end]

This layer contains the core business logic of the system and is responsible for processing requests from the UI layer. It acts as the controller of the system. Handle incoming requests, process them, and send responses back to the UI layer. It implement the core functionalities such as book management, catalog management, transaction processing, and reporting. The Application layer is located between the user interface layer and the data layer of the hierarchical model. Application layer is mainly used to realize the business functions of the management system, which is an important layer in the hierarchical model. This system comprised several modules, such as user registration, user login, literature resource search, literature resource reservation and so on.

4.1.3 Data Access Layer [Data Storage & Management]

This layer is responsible for data storage, retrieval, and management. Stores all the data related to books, members, and transactions. The data layer has access to external databases. It is the bottom layer of the whole system. By interacting with the database, we can complete operations such as query, insert, delete and modify. The data access layer is mainly responsible for data extraction and data editing of the application layer. The system includes user information, book content data, etc. The user's basic data comprises the basic data of general users and

librarians. The content data of the website is mainly about book ordering information, borrowing information, etc.

4.2 USE CASE DIAGRAM

The Use Case Diagram visually represents the interactions between users and the system showcasing different functionalities available to them.

The various actors are:

1. Librarian:

The librarian is responsible for managing the library system, including user and book management. Primary Responsibilities:

- Managing users (adding, deleting, and modifying users).
- Issuing and returning books.
- Adding and removing books.
- Managing book records (adding, editing, and deleting book details).

2. Users (Students & Staff)

Users are the library members who can access library services. They are divided into two categories: Students and Staff. Primary Responsibilities:

- Searching, reserving, returning and viewing books.
- Paying fines if applicable.
- Accessing e-books.
- Changing credentials (such as passwords).
- Registering as new users by filling up a registration form.

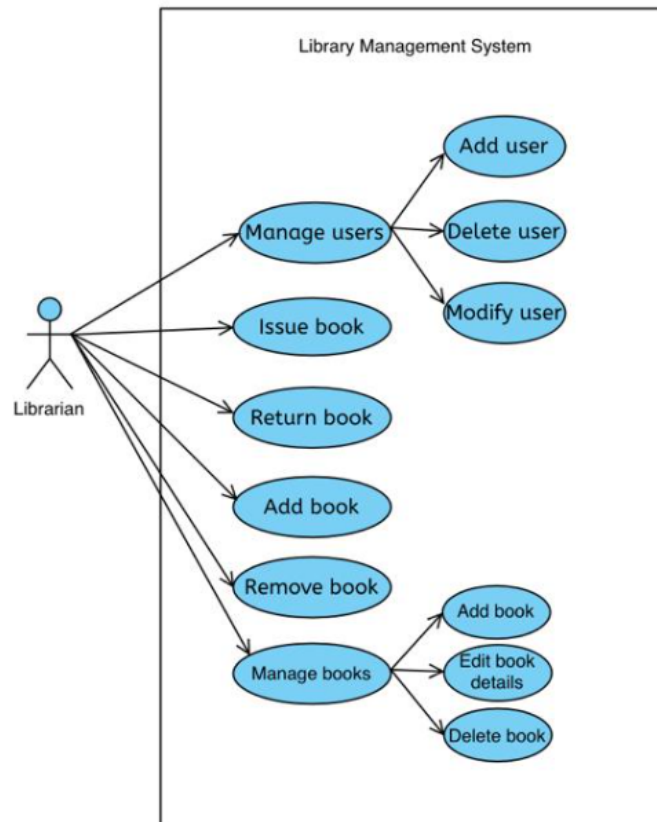


Figure 2: Use Case for Librarian

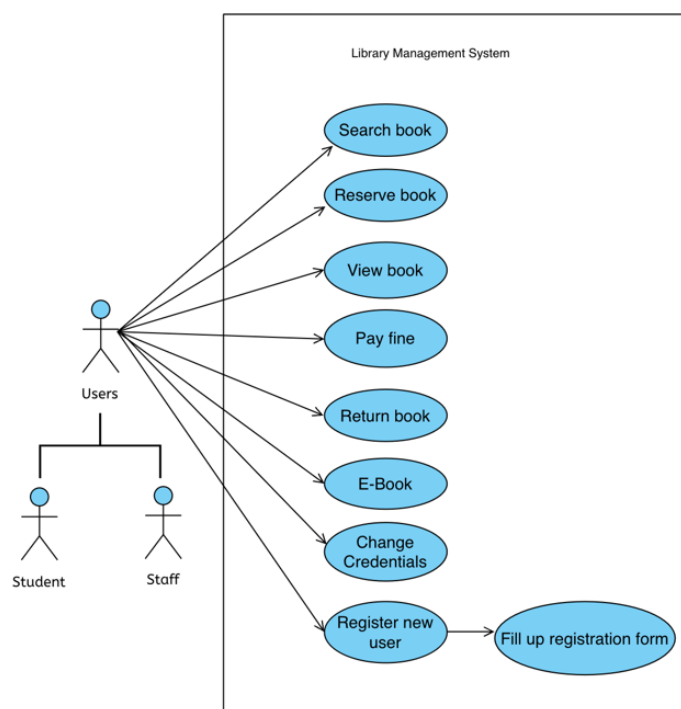


Figure 3: Use Case for User

4.3 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) is a structured analysis and design method. It is a visual tool to depict logic models and expresses data transformation in a system.

4.3.1 LEVEL-0 DFD

The diagram represents the high-level overview of how data flows between the main components of the system

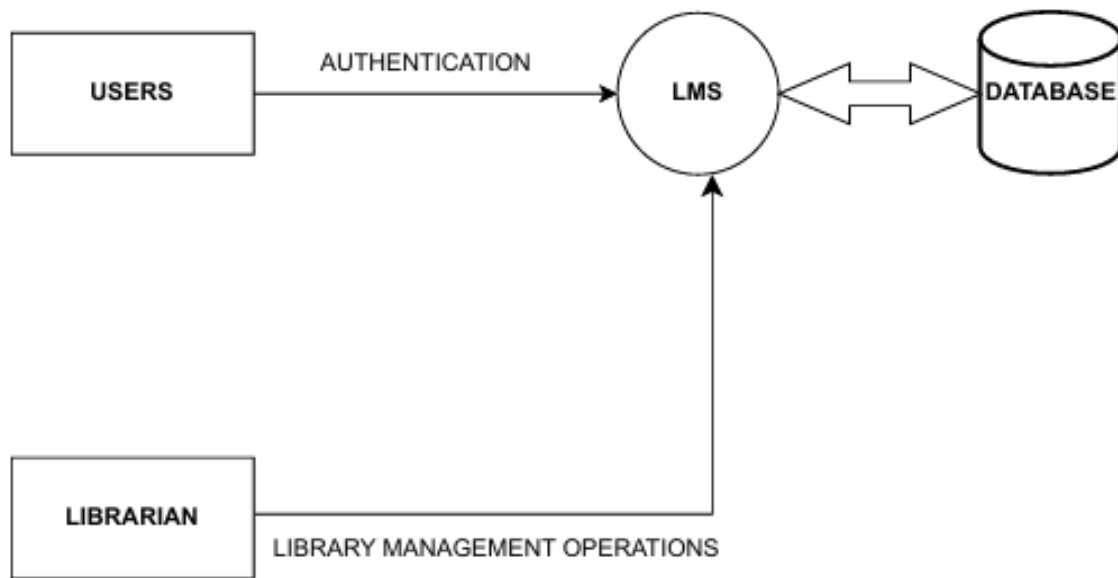


Figure 4: LEVEL-0 DFD

The diagram represents the high-level overview of the system, showing the interaction between external entities, LMS and Database

MAIN COMPONENTS :

1. USER

- The user interacts with the Library Management System(LMS)
- The user must log in to access the features of the LMS.
- Once authenticated, they can perform various actions like searching for books, borrowing books etc.

2. LIBRARIAN

- The librarian interacts with the system for library management operations, which includes adding, updating, or removing books, issuing books, handling returns, and managing users.

3. LIBRARY MANAGEMENT SYSTEM(LMS)

- Acts as the central processing unit that handles all interactions.
- It verifies user authentication and grants access to authorized users.
- It processes book-related transactions and librarian operations.

4.DATABASE

- Stores user credentials, book records, issued/returned books, and other essential library-related data.
- The LMS retrieves and updates records as needed.

DATA FLOW:

- Users → LMS: Sends authentication requests to log in.
- LMS → Database: Retrieves or verifies user credentials from the database.
- Librarian → LMS: Sends management operations such as book additions, deletions, and updates.
- LMS ↔ Database: Performs data retrieval and updates related to book and user records.

4.3.2 LEVEL-1 DFD

The Level 1 Data Flow Diagram (DFD) provides a more detailed breakdown of the system's processes and interactions

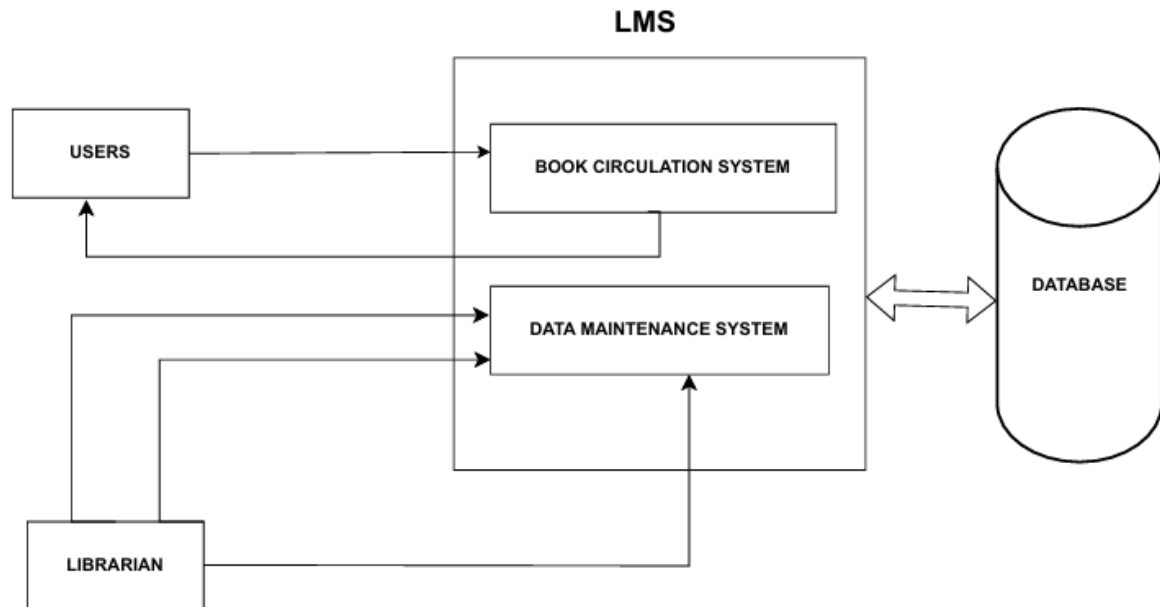


Figure 5: LEVEL-1 DFD

The Level 1 Data Flow Diagram (DFD) provides a detailed breakdown of the Library Management System (LMS), expanding on the high-level components from Level 0 DFD. It illustrates specific subsystems, data stores, and interactions between the entities (Users, Librarians, and LMS).

MAIN COMPONENTS :

1. BOOK CIRCULATION SYSTEM :

The Book Circulation System is responsible for managing book borrowing and returning processes.

- Authenticates users before granting access.
- Retrieves book details from the Book Info database.
- Verifies member eligibility by checking borrowing history from Member Info.
- Issues and returns books, updating the Loans database accordingly.
- Sends book due dates and return reminders to members.

2. DATA MAINTENANCE SYSTEM :

The Data Maintenance System handles all data updates related to books, members, and transactions.

- Manages book records (add, update, delete books).
- Stores and updates member details, including registration and borrowing history.
- Generates reports for the librarian on book circulation, overdue books, and system usage.
- Communicates with all databases to maintain data integrity.

OVERALL WORKFLOW:

- Users → Book Circulation System: Sends login credentials for authentication.
- Book Circulation System → Database: Retrieves book details based on user requests.
- Book Circulation System → Database: Retrieves member details to verify borrowing eligibility.
- Book Circulation System → Database: Updates loan information when books are issued or returned.
- Librarian → Data Maintenance System: Performs operations like adding books, updating member records, and generating reports.
- Data Maintenance System → Database: Updates loan and member records.
- All Systems ↔ Databases: Continuous data retrieval and updates.

DATABASE:

- Stores book details, availability status, and issue history.
- Stores user credentials and borrowing history.
- Stores information on book issues, due dates, and returns.
- Maintains details of book reservations made by members.

4.3.3 LEVEL-2 DFD

The Level 2 Data Flow Diagram (DFD) provides a detailed breakdown of the internal process within the system. It expands on Level 1 DFD.

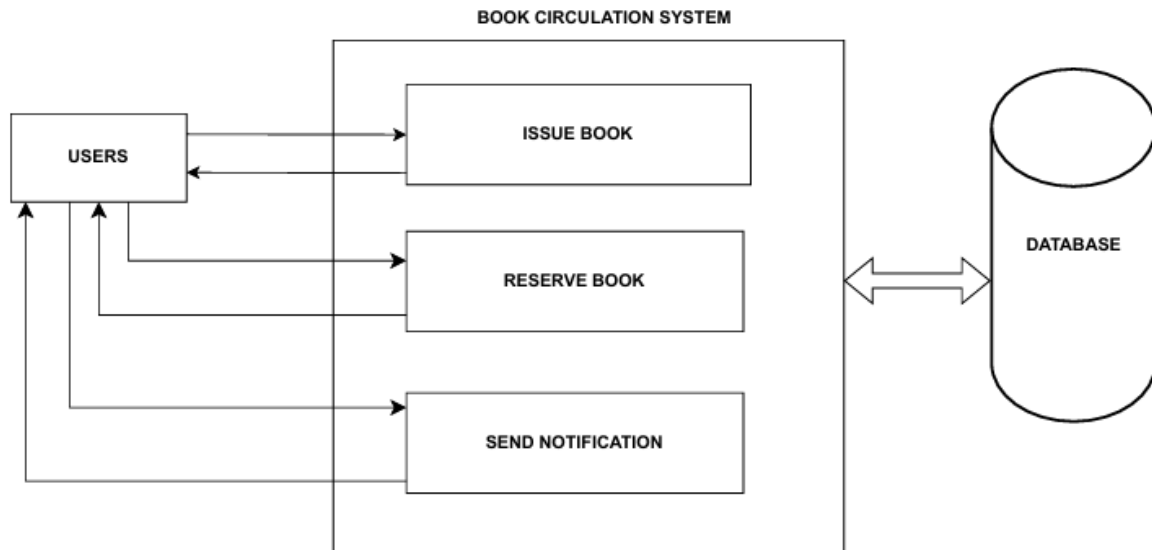


Figure 6: LEVEL-2 DFD 1

The Level 2 DFD 1 provides a detailed breakdown of the Book Circulation System by expanding its core processes. It highlights how different entities interact with specific subsystems to facilitate book issuance, reservations, and notifications.

MAIN COMPONENTS:

ISSUE BOOK:

- Handles the book lending process.
- Checks user credentials and book availability.
- Updates loan records with due dates.
- Sends book issue details to the Database.

RESERVE BOOK:

- Allows users to reserve books that are currently unavailable.
- Retrieves book and member information from the Database.

- Stores reservation details in the Database for future reference.

SEND NOTIFICATIONS:

- Notifies users about important library events.
- Sends reminders for book return due dates.
- Alerts users when a reserved book becomes available.
- Fetches loan and reservation details from the Database to generate notifications.

DATA FLOW:

- Users → Issue Book: Sends login credentials and book requests.
- Issue Book → Users: Confirms that the book has been issued successfully. Provides the due date and loan details to the user.
- Issue Book → Database: Retrieves book information from the Book Info database.
- Users → Reserve Book: Requests book reservation.
- Reserve Book → Users: If the reserved book becomes available, it is assigned to the user.
- Reserve Book → Database: Checks book availability. Stores reservation details in the Reservation System.
- Users → Send Notification: Users can request notifications for book availability or due date reminders.
- Send Notification → Users: Sends alerts about book due dates. Notifies users when reserved books are available.
- Database → Send Notification: Retrieves information about loan due dates and book reservations.

DATABASE SYSTEM:

- Stores book information, loan details, reservation records, and member details.
- Provides real-time data to different processes for accurate book circulation management.

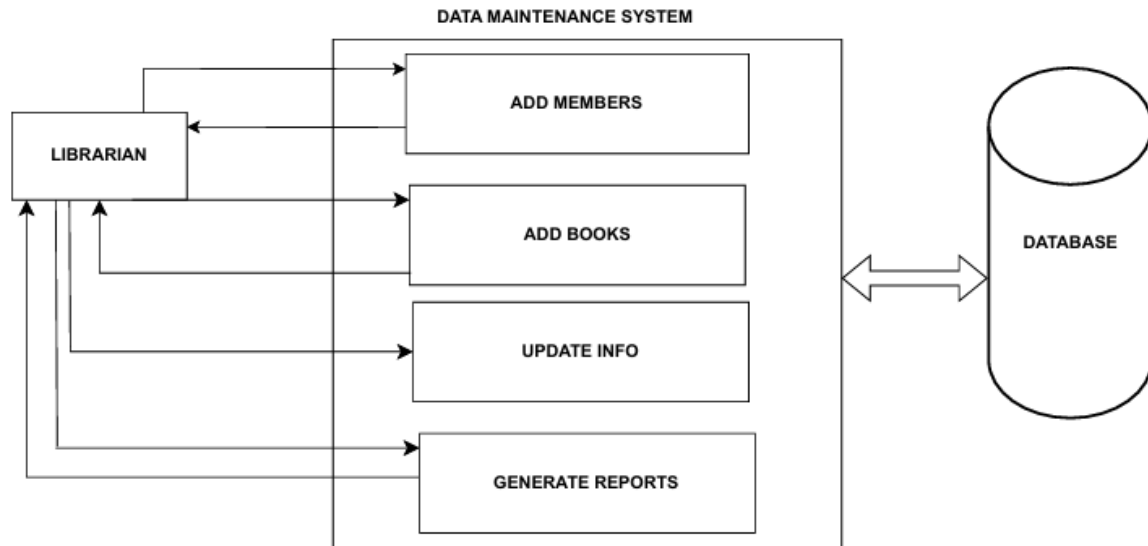


Figure 7: LEVEL-2 DFD 2

The Level 2 DFD 2 represents the Data Maintenance System, which is responsible for managing and updating library records efficiently. The core components of this system includes Add Members, Add Books, Update Info, Generate Reports.

MAIN COMPONENTS:

ADD MEMBERS:

- The librarian adds new members to the library system.
- Stores member information in the Member Info database.

ADD BOOKS:

- The librarian adds new books to the system.
- Updates the Book Info database with book details (title, author, ISBN, availability, etc.).

UPDATE INFO:

- The librarian updates book/member information as needed.
- Changes are reflected in the database.

GENERATE REPORTS:

- Generates detailed reports on book loans, reservations, and member activity.

- Fetches data from Book Info, Member Info, Loan, and Reservation databases.

DATA FLOW:

- Librarian → Add Members: Provides new member details.
- Add Members → Database: Stores new member information.
- Database → Add Members: Retrieves member records if needed.
- Librarian → Add Books: Enters book details.
- Add Books → Database: Stores book information.
- Database → Add Books: Retrieves book details if required.
- Librarian → Update Info: Requests modification of book/member data.
- Update Info → Database: Updates records.
- Database → Update Info: Retrieves existing information before modification.
- Librarian → Generate Reports: Requests reports on books, members, and transactions.
- Generate Reports → Database: Fetches required data.
- Database → Generate Reports: Supplies necessary information.
- Generate Reports → Librarian: Provides detailed reports.

DATA BASE SYSTEM:

It maintains book records by storing information about books, such as their availability and current status. Additionally, it tracks updates by keeping modification logs for both book and member records. Furthermore, the database generates reports by storing data related to book transactions, member activities, and system updates.

4.4 SEQUENCE DIAGRAM

A Sequence Diagram is a used to visualize the interaction between objects in a sequential order illustrating object interactions, message flows, and the sequence of operations

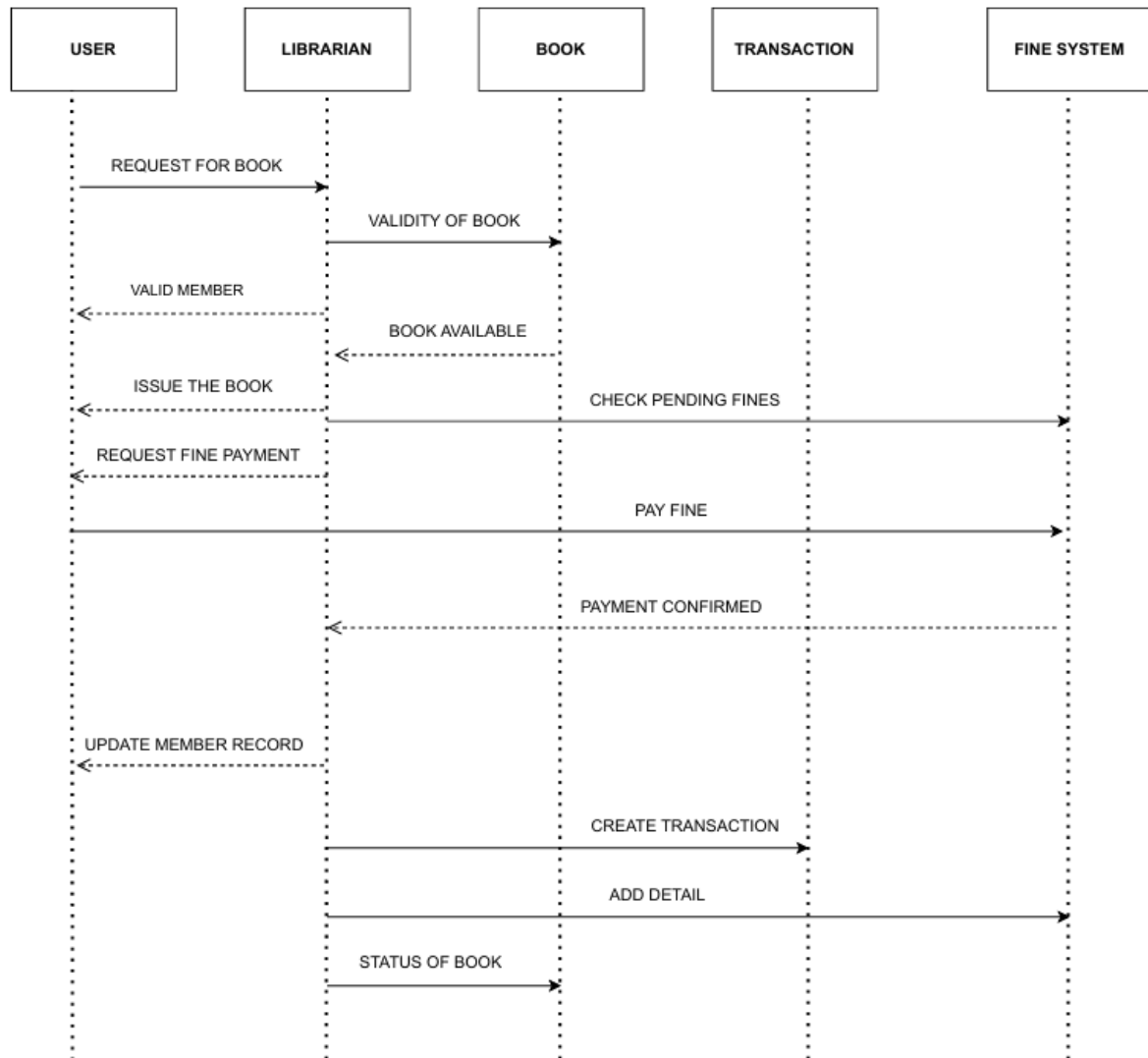


Figure 8: SEQUENCE DIAGRAM

CHAPTER 5

MODULE DESCRIPTION

The Department Library Management System is a comprehensive digital platform designed to streamline library operations, serving both administrators and users (students and staff) with robust, user-friendly features.

5.1 ADMINISTRATOR MODULE

The Administrator Module is designed for librarians or system administrators to manage books, users, and reports. This module provides complete control over the library's inventory, book issue/return process, and fine management. The administrator ensures that books are available, borrowed, and returned properly while keeping track of overdue books and fines.

MAIN FEATURES:

BOOK MANAGEMENT:

- The administrator can add, update, or remove books from the system.
- Each book has details like: Title, Author, ISBN (International Standard Book Number), Category (Science, Fiction, Technology, etc.), Availability Status (Available, Issued, Reserved, or Removed), Stored in the Book Database Table.

BOOK ISSUE AND RETURN :

- Books can be issued to students and staff by the librarian.
- While issuing a book: The system records the borrower's details (Name, User ID, Role), A due date is assigned based on library policies.
- When a book is returned: The system updates the loan record, Availability status is changed to Available.
- If a book is overdue, the system: Automatically calculates fines, Sends fine notifications to users.

RESERVATION SYSTEM :

- If a book is unavailable, users can reserve it.

- When the book is returned, the system notifies the user.
- The reservation will expire after a fixed time if not collected.

USER MANAGEMENT:

- The administrator can add, update, and remove users from the system.
- Users are classified as:
Students – Limited borrowing period, fewer books at a time,
Staff – Longer borrowing period, more books allowed.
- The system enforces role-based permissions.

REPORT GENERATION:

- Issued books and overdue books.
- Pending reservations.
- Fines collected and unpaid fines.
- Helps analyze library usage and book demand.

5.2 USER MODULE

The User Module is for students and staff, allowing them to search, reserve, borrow, and return books while managing their library activities.

MAIN FEATURES:

AUTHENTICATION SYSTEM :

- Users must log in before accessing library services.
- New users can register using their details.

BOOK SEARCH AND RESERVATION :

- Users can search books using: Title, Author, ISBN, Category.
- If a book is available, users can borrow it.

- If a book is unavailable, users can reserve it.

BORROWING AND RETURN MANAGEMENT :

- Users can view their borrowed books and due dates.
- If needed, users can request a renewal before the due date.
- Borrowing limits are implemented
- If a book is overdue, the system Calculates fines automatically.

PROFILE MANAGEMENT :

- Users can update their details such as: Name, Email, Contact Information.
- Users can also check their borrowing history.

References

- [1] Sarma, G.: ‘OPAC Module in Open Source Library Management Software: A Comparative Study’. *DESIDOC Journal of Library & Information Technology*, January 2016, pp. 56-61.
- [2] Făgărășan, C.: ‘Library Management: an Overview of Its Past, Present, and Future.’. *Revista Română de Biblioteconomie și Știința Informării*, September 2017, pp. 47-52.
- [3] eLibrary Software.: ‘What is an Online Library Management System?’ eLibrary Software, 2025.
- [4] eLibrary Software. “Features of Library Management System.” eLibrary Software, 2025.
- [5] Chinmay Kumar Roy. “Top 20 Common Problems in Any Library Management System Software.” LinkedIn, 2025.
- [6] Ying Li, ”Design of Library Archives Information Management Systems Based on Artificial Intelligence and Multimedia Technology,” **International Journal of Information Technologies and Systems Approach**, vol. 16, no. 3, pp. 1–17, 2023.